

CivicPulse Nexus

Cloud-Native Smart Governance & Citizen Services Management Platform

8 Weeks | 4 Milestones

1. Project Idea

CivicPulse Nexus is an enterprise-grade government and municipal management platform designed to digitize citizen services, grievance management, public administration, approvals, municipal operations, welfare schemes, permits, and governance workflows. Instead of AI agents, forecasting, and policy simulations, the platform focuses on Citizen Service Management, Grievance Redressal, Public Administration, Municipal Operations, Permit & License Management, Welfare Scheme Management, Budget & Resource Tracking, Workflow Automation, Cloud-Native Deployment, and DevOps Automation. The platform uses React 20 frontend with Java 25 and Spring Boot 4 backend. PostgreSQL database with Redis cache. Microservices include User Service, Citizen Service, Service Management Service, Grievance Service, Budget Service, and Reporting Service. Kafka enables event-driven architecture. Keycloak provides IAM. The system manages 2.4M citizens, processes 12.4K grievances monthly with 94% resolution, issues 847K certificates annually with a 2.4-day average approval, manages 247K welfare beneficiaries disbursing \$24.7M, and achieves 4.7/5 citizen satisfaction.

2. Emerging Skills

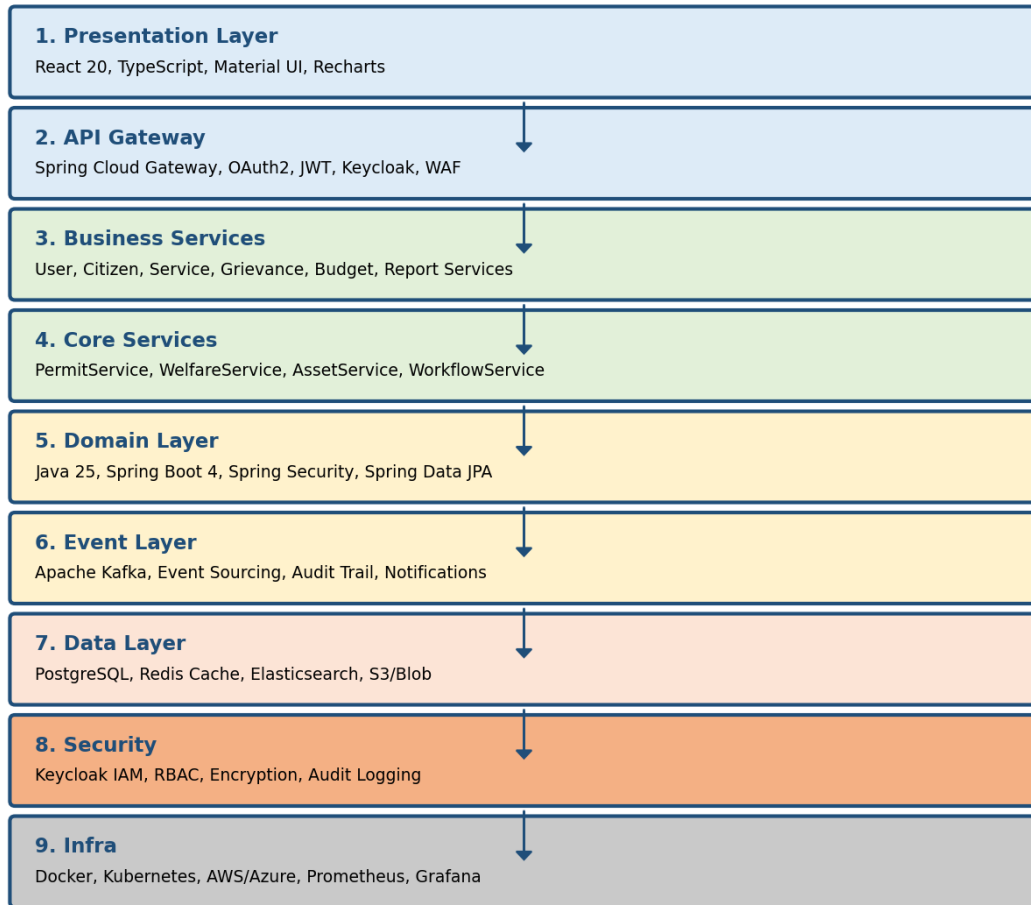
- Full Stack Java, Cloud, DevOps, Microservices, Event Driven Architecture, Banking Domain, Security

3. Technology Stack

Component	Technology
Frontend	React 20
Backend	Java 25, Spring Boot 4
Database	PostgreSQL
Cloud	AWS EC2, EKS, RDS, S3 OR Azure AKS, SQL, Blob
Skills	Full Stack Java, Cloud, DevOps, Microservices, Event Driven Architecture, Banking Domain, Security

4. System Architecture Diagrams

CivicPulse Nexus - 9-Layer Smart Governance Architecture



9-layer architecture with React 20 frontend, Spring Cloud Gateway, Java 25/Spring Boot 4 microservices, Kafka event streaming, and PostgreSQL data layer.

5. Project Working Diagram

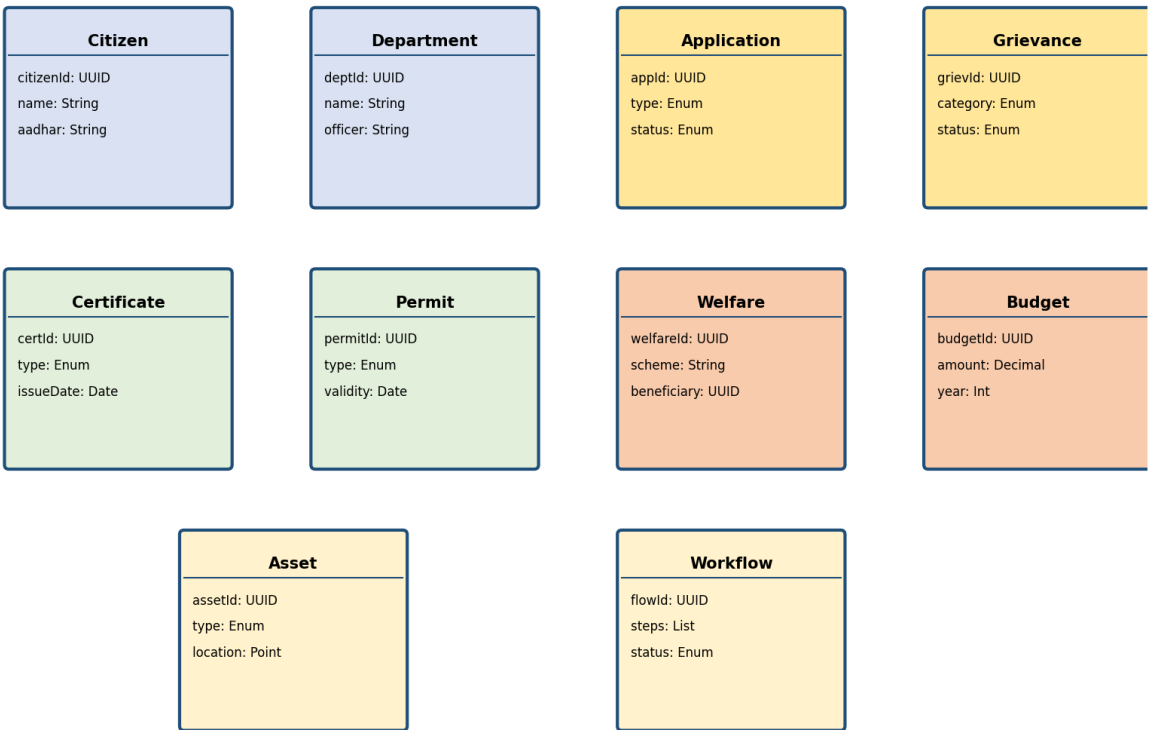
CivicPulse Nexus - Citizen Service Workflow



Citizen service workflow: Citizen Request → Service Application → Workflow Automation → Verification Approval → Certificate Generation → Budget Allocation → Analytics Dashboard.

6. Class Diagrams

CivicPulse Nexus - Class Diagram: Governance Domain



Core entities: Citizen, Department, Application, Grievance, Certificate, Permit, Welfare, Budget, Asset, Workflow.

7. Milestone-wise Deliverables & Screens

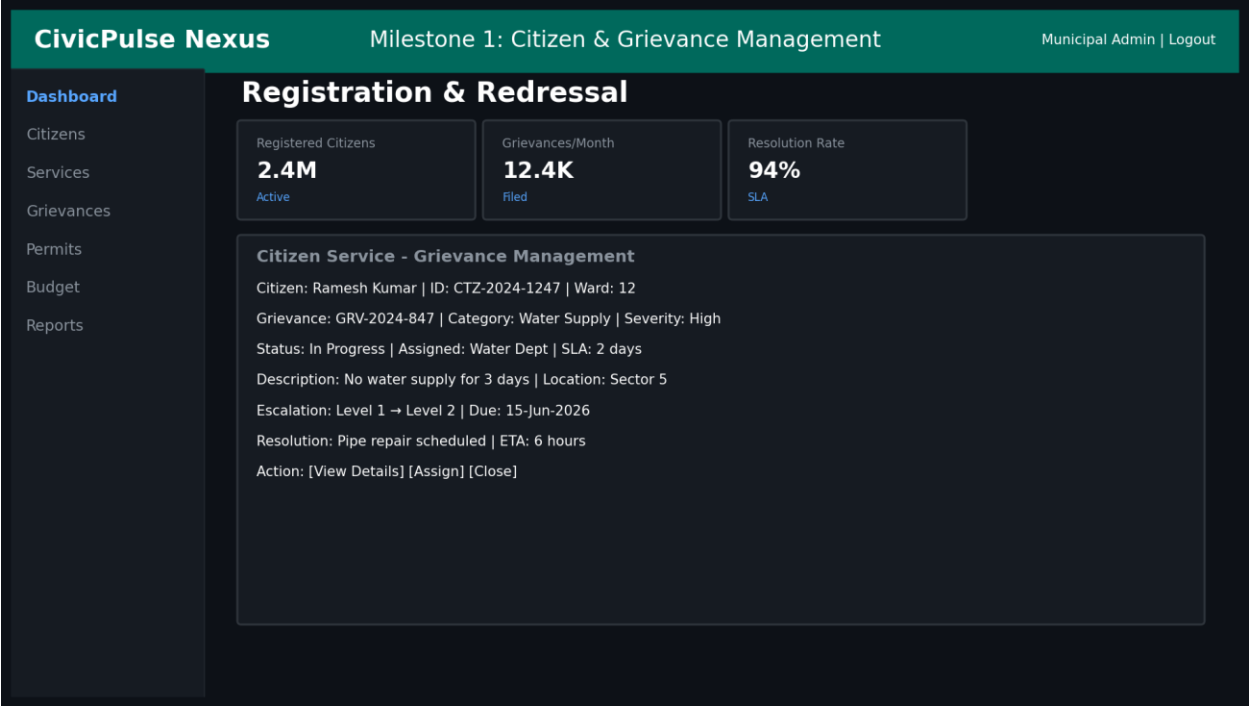
Milestone 1 (Weeks 1-2) – Citizen & Grievance Management

Implements citizen registration and grievance redressal. Manages 2.4M citizens. Processes 12.4K grievances monthly. Achieves 94% resolution rate. Tracks complaint workflow from registration to closure. Assigns to departments. Monitors SLA compliance.

Key Modules

- Citizen Service
- Grievance Service
- Registration
- Complaint Tracking
- Escalation Workflow
- SLA Monitoring

Expected Output Screen



Citizen Service: Ramesh Kumar, Ward 12. Grievance: Water Supply. Status: In Progress. Assigned: Water Dept. SLA: 2 days.

Validation Screens

- Citizen registration
- Grievance registration

- Assignment workflow
- Escalation rules
- SLA tracking
- Resolution validation

Milestone 2 (Weeks 3-4) – Certificate & Permit Management

Deploys certificate and permit management. Processes 24.7K applications monthly. Achieves 2.4 day average approval. Issues 847K certificates annually. Manages birth, death, income, residence certificates and trade licenses. Implements approval workflow.

Key Modules

- Service Management Service
- Certificate Generation
- Permit Management
- License Tracking
- Approval Workflow
- Document Verification

Expected Output Screen

CivicPulse Nexus Milestone 2: Certificate & Permit Management Municipal Admin | Logout

Service Delivery & Approvals

Applications/Month 24.7K Processed	Avg Time 2.4 days Approval	Certificates 847K Issued
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Service Management - Certificate Generation

Application: APP-2024-1247 | Type: Birth Certificate

Applicant: Priya Sharma | Child: Aarav | DOB: 15-May-2026

Documents: Hospital Record ✓ Aadhar ✓ | Verified

Workflow: Submitted → Verified → Approved → Generated

Certificate: BC-2024-1247 | Issued: 15-Jun-2026

Status: Approved | Download: Available | Digital signed

Action: [View App] [Approve] [Download]

Service Management: Birth Certificate APP-2024-1247. Applicant: Priya Sharma. Verified. Approved. Certificate BC-2024-1247 issued.

Validation Screens

- Application submission

- Document verification
- Approval workflow
- Certificate generation
- Digital signing
- Download tracking

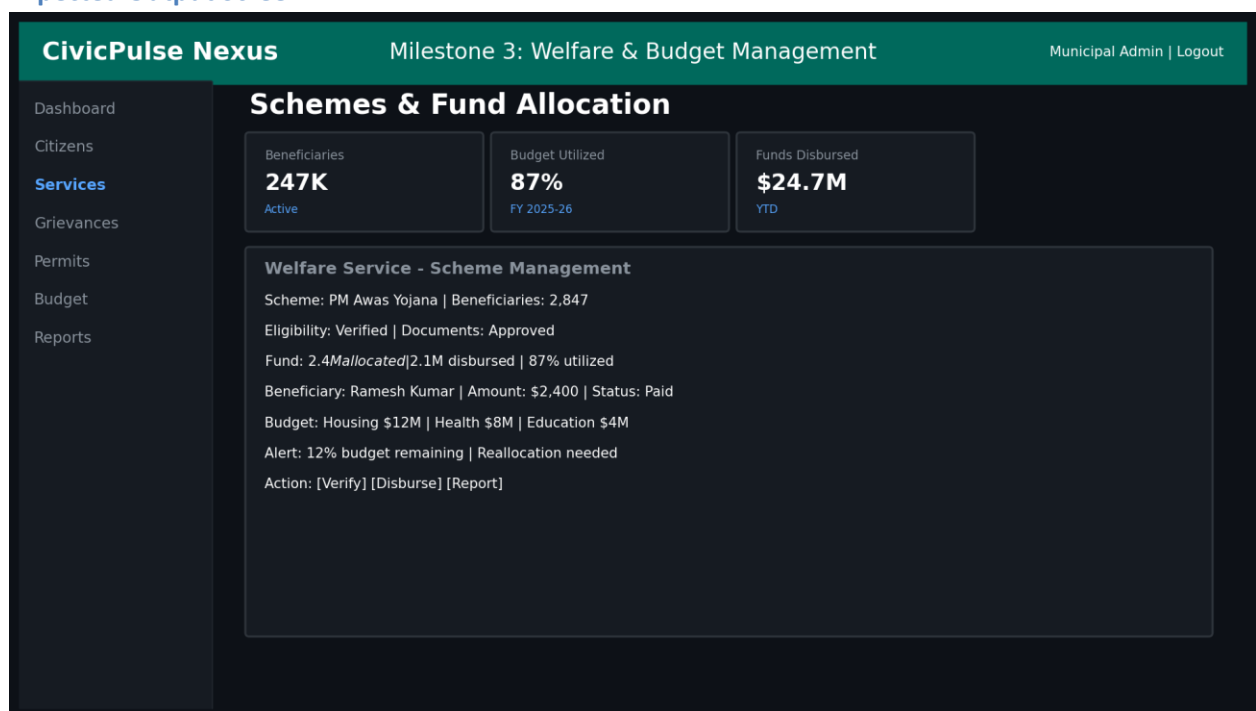
Milestone 3 (Weeks 5-6) – Welfare & Budget Management

Implements welfare scheme and budget management. Manages 247K beneficiaries. Disburses \$24.7M in funds. Achieves 87% budget utilization. Tracks eligibility and fund distribution. Monitors department budgets. Generates utilization reports.

Key Modules

- Welfare Service
- Scheme Management
- Beneficiary Management
- Budget Service
- Fund Distribution
- Expense Tracking

Expected Output Screen



Welfare Service: PM Awas Yojana. 2,847 beneficiaries. \$2.4M allocated, \$2.1M disbursed. 87% utilized. Ramesh Kumar paid \$2,400.

Validation Screens

- Eligibility verification
- Beneficiary management
- Fund disbursement
- Budget tracking
- Utilization monitoring
- Compliance check

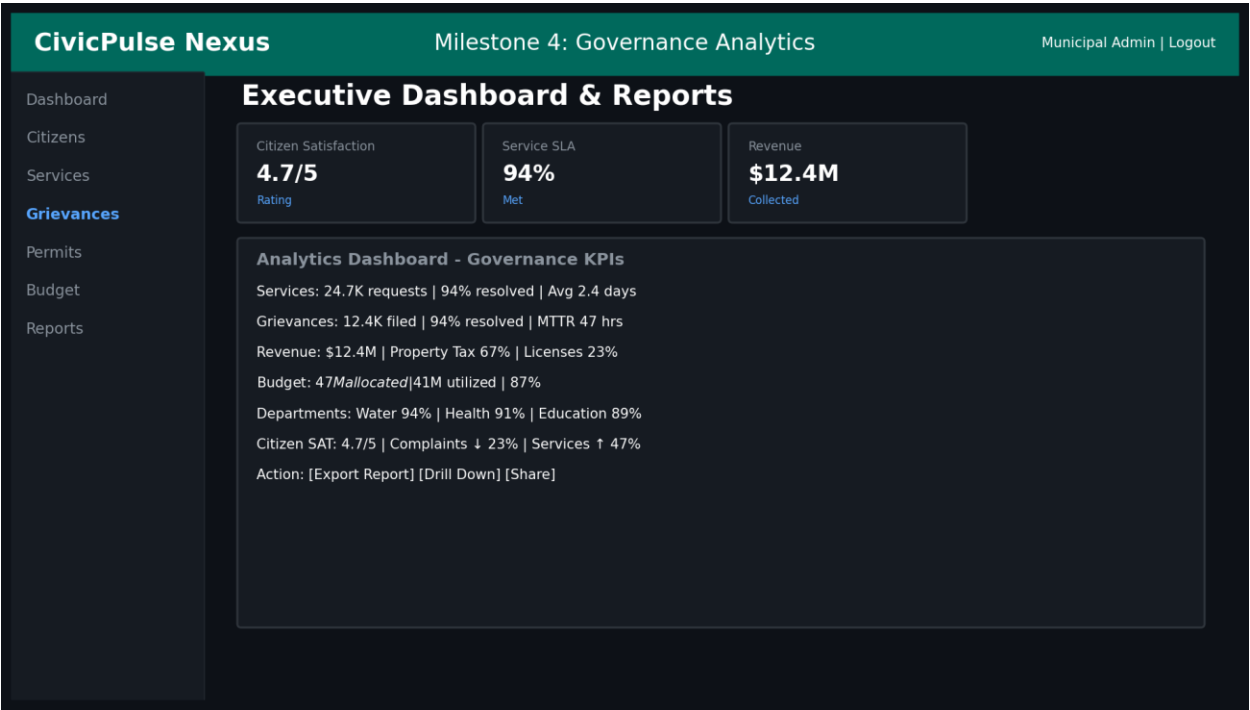
Milestone 4 (Weeks 7-8) – Governance Analytics

Deploys an executive governance dashboard. Tracks citizen satisfaction 4.7/5. Monitors 94% service SLA. Reports \$12.4M revenue collected. Analyzes department performance. Provides grievance, service, and budget reports. Supports compliance audits.

Key Modules

- Reporting Service
- Analytics Dashboard
- Citizen Reports
- Grievance Reports
- Revenue Reports
- Performance Reports

Expected Output Screen

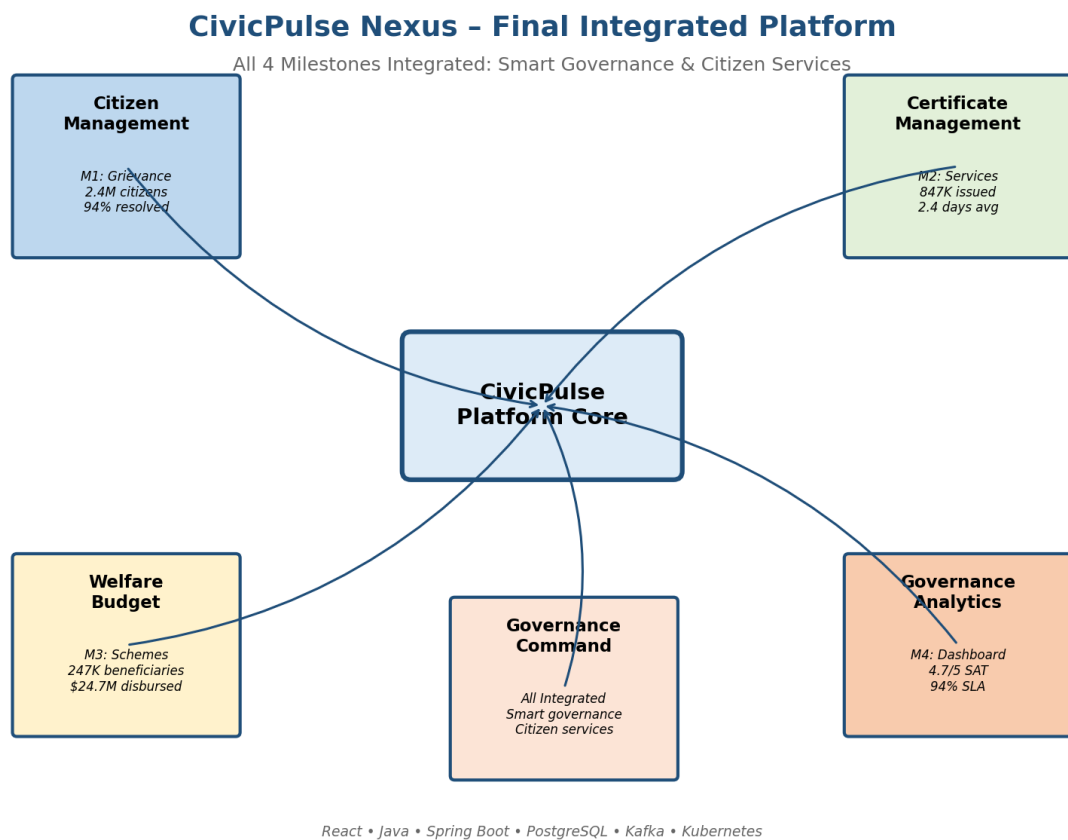


Analytics Dashboard: 24.7K requests, 94% resolved. Revenue \$12.4M. Budget 87% utilized. Citizen SAT 4.7/5. Complaints ↓ 23%.

Validation Screens

- Service metrics
- Grievance analytics
- Revenue tracking
- Budget utilization
- Department performance
- Citizen satisfaction

8. Master Application Screens – Final Integrated



Final integrated master screens after all 4 milestones:

- Citizen Management: M1 – Grievance, 2.4M citizens, 94% resolved, 12.4K/month
- Certificate Management: M2 – Services, 847K issued, 2.4 days avg, 24.7K apps/month
- Welfare Budget: M3 – Schemes, 247K beneficiaries, \$24.7M disbursed, 87% utilized
- Governance Analytics: M4 – Dashboard, 4.7/5 SAT, 94% SLA, \$12.4M revenue
- Governance Command: All integrated – Smart governance, citizen services, public administration

9. Validation Screens Framework

All screens enforce: Citizen data validation, Application workflow tracking, Service SLA monitoring, RBAC by admin/commissioner/officer via Keycloak, and comprehensive audit logging for all governance actions. Operations show request status + workflow stage + SLA compliance. Critical approvals require authorization. Kafka ensures event ordering. All activities are logged immutably. Citizens validated. Services tracked. Budgets audited.